EWB EDUTECH PRIVATE LIMITED

“STUDENT PERFORMANCE ANALYTICS SYSTEM”

PROJECT REPORT

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Course
Python With AI

Project type
Python Data Analysis

Submitted as part of the Python With AI project assignment

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1. Objective

The objective of this project is to analyse student marks stored in a CSV file and present the results in a clear, organised report. Calculating totals, averages, grades, and class level statistics manually can be time consuming and may lead to calculation errors. This program automates these calculations using Python, Pandas, and NumPy so that student level and subject level performance can be reviewed together.

The project focuses on fundamental Python programming and data analysis rather than advanced machine learning. It demonstrates how a structured dataset can be validated, processed, summarised, and exported for further use.

2. Technologies Used

Technology	Role in the Project
Python	Implements program logic and reusable functions.
Pandas	Loads, organises, filters, sorts, analyses, and exports tabular data.
NumPy	Supports numerical calculations and attendance related labels.
CSV	Stores the input dataset and generated result files.
unittest	Checks grade boundaries and result calculations.

3. Dataset Description

The program reads students.csv, which contains 22 fictional student records. Each record includes identifying fields, department information, marks in four subjects, and an attendance percentage.

Column / Group	Description
student_id	Unique identifier for a student.
name	Student name used in reports and sorting.
department	Department associated with the student record.
Python, Mathematics, Statistics, Data Analysis	Subject marks, expected to be between 0 and 100.
attendance_percent	Attendance percentage, expected to be between 0 and 100.

The sample data is fictional and intended for demonstration. It includes students with marks below the pass threshold in at least one subject, allowing both passing and failing outcomes to be demonstrated. Input checks are designed to reject missing required columns, empty data, blank numeric values, and marks or attendance outside the accepted range.

4. Implementation

Step	Implementation Details
Read the dataset	main.py loads students.csv into a Pandas DataFrame using pd.read_csv().
Validate input	load_students() checks required columns, non-empty data, and numeric marks/attendance within 0–100.
Calculate totals and averages	prepare_results() copies the data, calculates each student's total and mean marks, and rounds averages to two decimal places with NumPy.
Assign grades	grade_from_average() applies the documented average based grade boundaries.
Pass/fail and attendance	A student passes only when every subject mark is at least 35. Attendance of 75% or more is labelled Eligible.
Summarise performance	class_summary() calculates student count, class average, highest and lowest averages, pass/fail counts, and subject averages.
Rank students	top_students() sorts by average in descending order and uses name to order tied averages.
Display and export	show_report() prints summaries and results. save_outputs() writes student results and subject averages to CSV files.

Grade Boundaries

Average Marks	Grade
90 and above	A+
80 to below 90	A
70 to below 80	B
60 to below 70	C
50 to below 60	D
Below 50	F

The grade is based on the overall average, while pass/fail status depends on the minimum mark in every subject. These are separate rules and may produce different labels for a student.

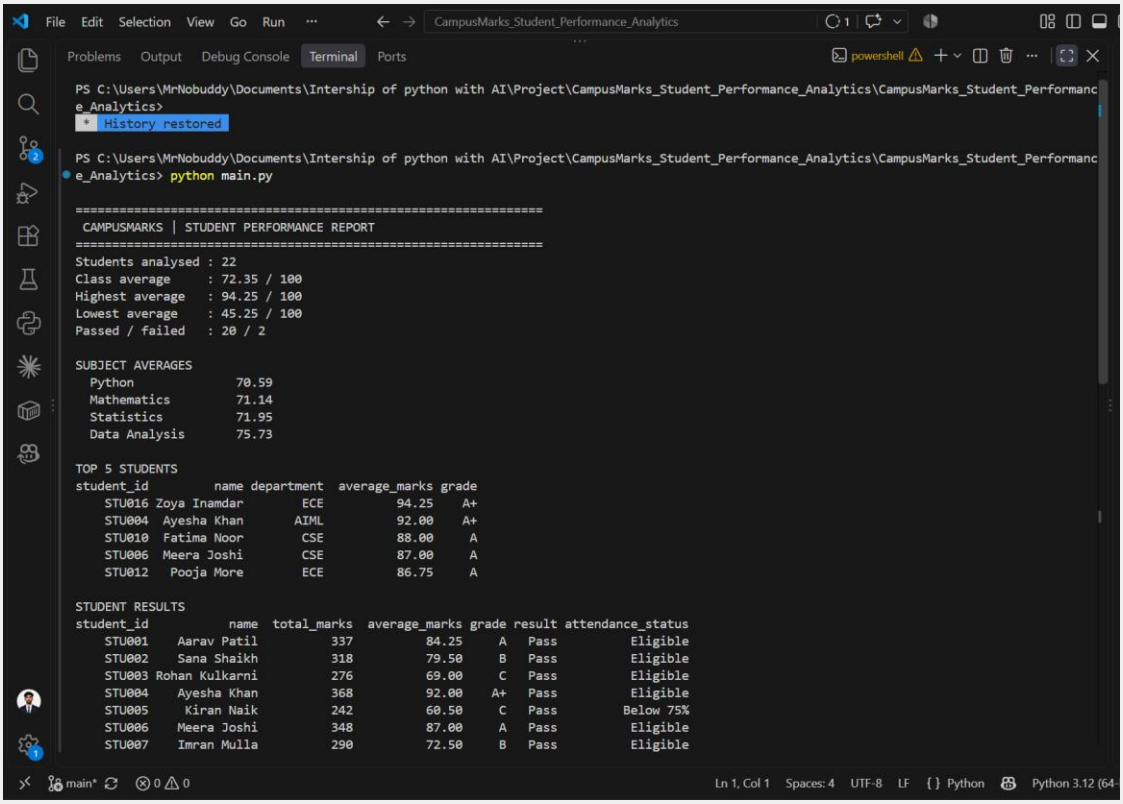
5. Key Features

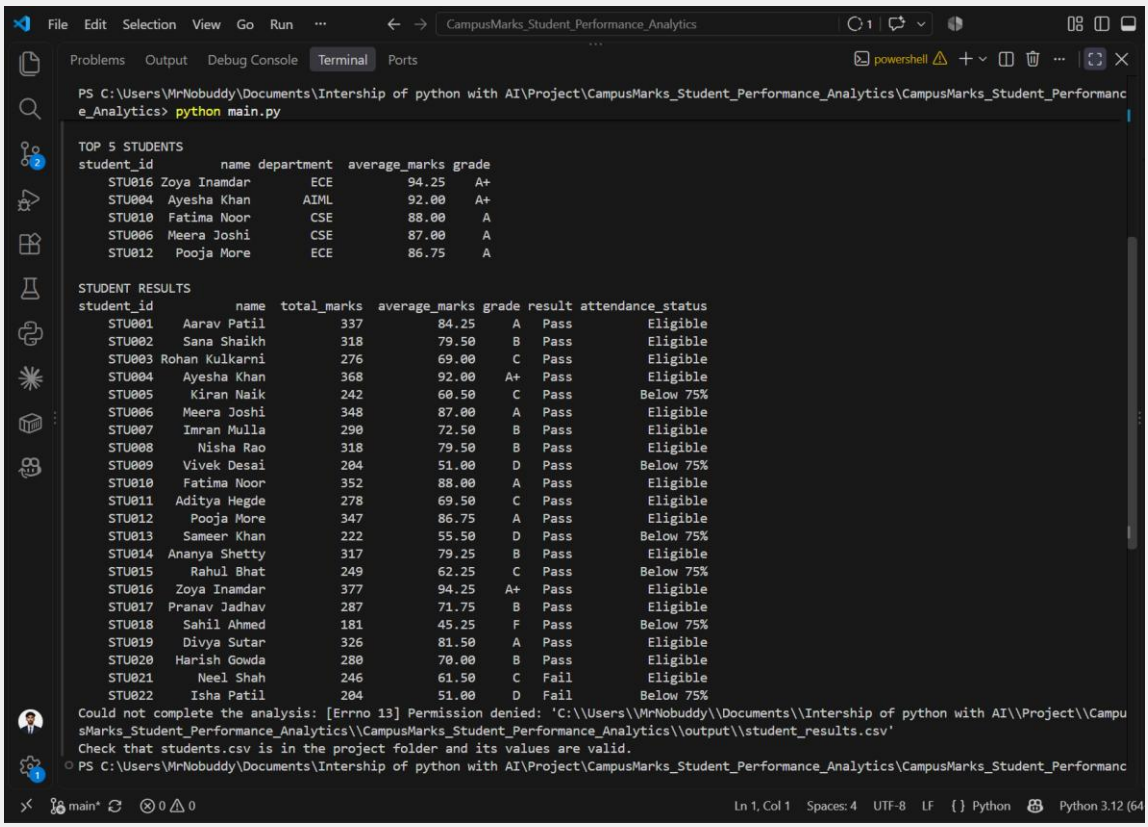
- Reads student records from a CSV file and validates the input.
- Calculates total and average marks for each student.
- Assigns grades using documented average mark boundaries.
- Identifies pass/fail status using a minimum of 35 marks in every subject.
- Reports class average, highest and lowest averages, and pass/fail counts.
- Calculates subject-wise average marks.
- Displays the top five students based on average marks.
- Labels attendance of 75% or more as Eligible.
- Exports detailed student results and subject averages as CSV files.
- Includes unit tests for grade boundaries and calculated results.

6. Output / Screenshots

The program displays a summary in the terminal and saves the generated files in the output folder. Add a genuine screenshot captured after running the program locally.

Generated File	Contents
output/student_results.csv	Student details, totals, averages, grades, pass/fail status, and attendance label.
output/subject_averages.csv	Average marks for each subject.





7. Final Outcome

According to the supplied project results, the program analysed 22 students. The reported class average was 72.35, the highest student average was 94.25, and the lowest was 45.25. The summary recorded 20 passes and 2 failures. Data Analysis had the highest subject average, at 75.73.

The grade and pass/fail checks are independent: the grade is determined by the overall average, while passing requires at least 35 marks in every subject. Consequently, the overall grade should not be treated as a replacement for the subject-wise pass condition.

The supplied project notes state that the program ran successfully and that both tests in test_project.py passed. Recheck these results after changing the source code or dataset.

8. Challenges and Learning

One important implementation concern is data quality. Missing columns, blank marks, and values outside the allowed range can affect calculation reliability. Input validation helps identify these issues before analysis.

Separating data loading, calculations, summary generation, display, and file export into different functions makes the program easier to understand and test. The project also provides practical experience with DataFrames, numerical operations, sorting, conditional logic, loops, CSV files, and unit testing.

9. GitHub Repository

Project repository:https://github.com/MdZubair-22/Student-Performance-Analytics-System